

A Literature-Implied Negative Answer to the Argyros–Haydon $\mathcal{L}_\infty$ Strictly Singular Question

Math discovery packet

June 29, 2026

1 Status

This is a lightweight *literature-implied answer*, not a new construction. The source paper is S. A. Argyros and R. G. Haydon, *A hereditarily indecomposable $\mathcal{L}_\infty$ -space that solves the scalar-plus-compact problem*, arXiv:0903.3921. In the concluding open problems, after asking whether an isomorphic predual of ℓ_1 with scalar-plus-strictly-singular operators must also have the scalar-plus-compact property, the paper asks the more general problem:

Let X be a $\mathcal{L}_\infty$ -space. Is every strictly singular operator on X weakly compact?

The supporting paper is A. Aviles, F. Cabello Sanchez, J. M. F. Castillo, M. Gonzalez and Y. Moreno, *On separably injective Banach spaces*, arXiv:1103.6064.

2 Identification

Section 2 of arXiv:1103.6064 discusses Pelczynski's property (V). It states that not all $\mathcal{L}_\infty$ -spaces have property (V), and among its examples records the space Ω of Cabello Sanchez–Castillo–Kalton–Yost in a twisted sum

$$0 \longrightarrow C[0, 1] \xrightarrow{i} \Omega \xrightarrow{q} c_0 \longrightarrow 0$$

with q a strictly singular quotient map. Thus Ω is an $\mathcal{L}_\infty$ -space and $q : \Omega \rightarrow c_0$ is strictly singular.

Since $C[0, 1]$ contains a subspace isomorphic to c_0 , choose an isomorphic embedding $e : c_0 \rightarrow C[0, 1]$, and put

$$J = i \circ e : c_0 \rightarrow \Omega, \quad T = Jq : \Omega \rightarrow \Omega.$$

3 The Counterexample Operator

The operator T is strictly singular. Indeed, q is strictly singular and the class of strictly singular operators is a two-sided operator ideal, so post-composition by the bounded map J preserves strict singularity.

The operator T is not weakly compact. Since q is a quotient map, the open mapping theorem gives a constant $C < \infty$ such that every $y \in B_{c_0}$ has some $x \in CB_\Omega$ with $qx = y$. Therefore

$$J(B_{c_0}) \subseteq T(CB_\Omega).$$

If T were weakly compact, then $T(CB_\Omega)$ would be relatively weakly compact, hence so would $J(B_{c_0})$. But J is an isomorphic embedding of c_0 , and the weak topology on $J(c_0)$ is inherited from Ω . Thus relative weak compactness of $J(B_{c_0})$ would make the unit ball of c_0 weakly compact, contradicting the non-reflexivity of c_0 .

Consequently the $\mathcal{L}_\infty$ -space Ω admits a strictly singular operator on itself that is not weakly compact. This gives a negative answer to the Argyros–Haydon problem as stated.

4 Scope

This does not solve the separate reflexive scalar-plus-compact problem from the same concluding section of arXiv:0903.3921. It also does not decide the more special question about isomorphic preduals of ℓ_1 with the scalar-plus-strictly-singular property; the counterexample above is supplied by a different $\mathcal{L}_\infty$ -space.

The implication appears to be agent-identified. A local search of arXiv:1103.6064 found the twisted-sum example and found no explicit citation to arXiv:0903.3921 or to the Argyros–Haydon scalar-plus-compact problem in the supporting passage.

5 References

- S. A. Argyros and R. G. Haydon, *A hereditarily indecomposable $\mathcal{L}_\infty$ -space that solves the scalar-plus-compact problem*, arXiv:0903.3921.
- A. Aviles, F. Cabello Sanchez, J. M. F. Castillo, M. Gonzalez and Y. Moreno, *On separably injective Banach spaces*, arXiv:1103.6064.
- F. Cabello Sanchez, J. M. F. Castillo, N. J. Kalton and D. T. Yost, *Twisted sums with $C(K)$ -spaces*, Trans. Amer. Math. Soc. 355 (2003), 4523–4541.